

Primitive Coordinate Heights and a Flat-Pressure Obstruction for the Area-Preserving Hénon Map

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Abstract

For the area-preserving Hénon map $H_6(q, p) = (1 - 6q^2 - p, q)$, previous all-period results make every odd primitive mixed-axis divisor reduced, effective, and totally real. This invites a pressure formed by weighting each primitive root with its absolute logarithmic Weil height. We prove that this pressure is necessarily flat. After the integral scaling $x = 6q$, every primitive root is an algebraic integer and all its conjugates are real periodic coordinates. A maximum principle for the cyclic recurrence gives the sharp uniform bound $|x| \leq 1 + \sqrt{7}$, hence $h(x) \leq \log(1 + \sqrt{7})$ at every period. The height-weighted root count therefore differs from its unweighted degree by only a period-independent factor. Its pressure is exactly $\frac{1}{2} \log 2$ for every fixed real height parameter. Exact finite certificates reconstruct the scaled primitive polynomials through odd period 11 and independently verify the root bound through period 9. The result is a normalization obstruction, not an arithmetic promotion: a nontrivial next pressure must use an extensive observable such as period times height, packet Mahler height, or discriminant/ramification height.

1 Introduction

Arithmetic dynamics routinely uses absolute Weil height to measure the complexity of algebraic points; see, for example, Silverman [4]. In a periodic-orbit pressure, however, the relevant weight must also be extensive in the orbit clock. These two roles need not agree.

This paper isolates that mismatch for the frozen map

$$H_6(q, p) = (1 - 6q^2 - p, q). \quad (1)$$

Earlier work combines the hyperbolic plateau of Arai [1], the large-parameter full horseshoe of Devaney and Nitecki [2], and the complex fixed-point count of Friedland and Milnor [3]. At the parameter in (1), every complex periodic point is a distinct real hyperbolic point. Consequently every odd mixed-axis primitive divisor is squarefree, effective, and totally real.

The next proposed object was

$$\mathcal{P}_{\text{coord}}(s) = \limsup_{\substack{n \rightarrow \infty \\ n \text{ modd}}} \frac{1}{n} \log \sum_{\tilde{\Psi}_n(\alpha)=0} e^{-sh(\alpha)}, \quad (2)$$

where $\tilde{\Psi}_n$ is the integrally scaled primitive polynomial. Our main result shows that (2) contains no new pressure information: it is identically the unweighted half entropy.

2 Frozen primitive divisor and integral model

Put $x = 6q$ and $y = 6p$. The conjugate map is

$$(x, y) \mapsto (6 - x^2 - y, x), \quad (3)$$

and a cyclic period- n orbit satisfies

$$x_{j+1} = 6 - x_j^2 - x_{j-1}. \quad (4)$$

The cyclic equations are monic in x_j^2 with pairwise coprime leading monomials. Their quotient algebra is therefore finite free over $\mathbb{Z}$, with squarefree standard-monomial basis. In particular every periodic coordinate is an algebraic integer.

For odd n , let $\Psi_n(X) \in \mathbb{Q}[X]$ be the monic primitive mixed-axis quotient established in the preceding reflection-divisor work, and write $D_n = \deg \Psi_n$. Define

$$\tilde{\Psi}_n(T) = 6^{D_n} \Psi_n(T/6). \quad (5)$$

Every root of (5) is an algebraic integer. Since the polynomial is monic with rational coefficients, its coefficients are rational algebraic integers and hence ordinary integers. The previous algebraic-exhaustion theorem also makes it squarefree and totally real.

The exact primitive degree is

$$D_n = \sum_{d|n} \mu(n/d) 2^{(d+1)/2} = 2^{(n+1)/2} + O(n^{n/6+1/2}). \quad (6)$$

Thus

$$\lim_{\substack{n \rightarrow \infty \\ n \text{ mod } d}} \frac{1}{n} \log D_n = \frac{1}{2} \log 2. \quad (7)$$

3 A sharp uniform bound for every conjugate

Lemma 3.1 (Cyclic maximum principle). *Every coordinate on every real periodic orbit of (3) satisfies*

$$|x_j| \leq B := 1 + \sqrt{7}. \quad (8)$$

The constant is sharp.

Proof. Choose j with $M = |x_j|$ maximal along the orbit. Equation (4) gives

$$M^2 = |6 - x_{j-1} - x_{j+1}| \leq 6 + 2M.$$

Hence $M^2 - 2M - 6 \leq 0$, whose nonnegative endpoint is $1 + \sqrt{7}$. For sharpness, the fixed-coordinate equation is $x^2 + 2x - 6 = 0$; its negative root is $-1 - \sqrt{7}$. $\square$

Corollary 3.2 (All-conjugate bound). *If α is any root of $\tilde{\Psi}_n$, every Galois conjugate $\sigma(\alpha)$ is real and satisfies $|\sigma(\alpha)| \leq B$.*

Proof. The minimal polynomial of α divides $\tilde{\Psi}_n$. Hence every conjugate is another root of the same primitive divisor. Algebraic exhaustion identifies every such root with a real periodic coordinate, so Lemma 3.1 applies. $\square$

4 Uniform Weil height

Let $h(\alpha)$ denote the absolute logarithmic Weil height. If α is an algebraic integer, its nonarchimedean contributions vanish, and

$$h(\alpha) = \frac{1}{[K : \mathbb{Q}]} \sum_{\sigma: K \hookrightarrow \mathbb{C}} \log \max\{1, |\sigma(\alpha)|\} \quad (9)$$

for any number field K containing it.

Theorem 4.1 (Uniform primitive-coordinate height). *For every odd period and every root of the scaled primitive divisor,*

$$0 \leq h(\alpha) \leq C := \log(1 + \sqrt{7}). \quad (10)$$

Proof. Combine algebraic integrality with Corollary 3.2 in (9). $\square$

The upper bound is deliberately uniform rather than asymptotic. It does not assert that the finite heights converge, nor that primitive polynomials stay irreducible.

5 The flat-pressure theorem

For $s \in \mathbb{R}$, define the distinct-root partition function

$$Z_n(s) = \sum_{\tilde{\Psi}_n(\alpha)=0} e^{-sh(\alpha)}. \quad (11)$$

Squarefreeness ensures that each primitive root is counted once.

Theorem 5.1 (Flat ordinary height pressure). *For every fixed real s , the limit in (2) exists and*

$$\mathcal{P}_{\text{coord}}(s) = \frac{1}{2} \log 2. \quad (12)$$

Proof. Theorem 4.1 gives the uniform sandwich

$$e^{-|s|C} D_n \leq Z_n(s) \leq e^{|s|C} D_n. \quad (13)$$

After taking logarithms and dividing by n , the two errors are bounded by $|s|C/n$. Equation (7) proves (12). $\square$

Corollary 5.2 (Normalization obstruction). *The ordinary individual coordinate height cannot change the exponential primitive-root growth rate at any fixed s . In particular it cannot, by itself, generate a new pressure singularity or a nonconstant pressure curve.*

Corollary 5.3 (Fixed-rescaling invariance). *Let $c \neq 0$ be a fixed algebraic number and replace every root α in (11) by $c\alpha$. The resulting pressure is still $\frac{1}{2} \log 2$ for every fixed real s .*

Proof. The standard height inequality $h(c\alpha) \leq h(\alpha) + h(c)$ gives a period-independent upper bound. The same sandwich proof applies. $\square$

This conclusion is scoped to the observable in (11). The weights $e^{-snh(\alpha)}$, a packet Mahler mass, and discriminant or ramification heights are different objects.

Table 1: Finite primitive scaled-coordinate diagnostics. Heights are factor heights; all displayed finite polynomials are irreducible, a fact not promoted beyond the table.

n	D_n	h (diagnostic)	$\max \alpha $ (diagnostic)
1	2	0.895879734614	3.645751311065
3	2	0.693147180560	3.236067977500
5	6	0.592642696020	3.353191857132
7	14	0.648968099587	3.605741786740
9	28	0.660700003269	3.640173216825
11	62	0.670373466816	3.644971653259

6 Finite exact and numerical certificate

The executable package reconstructs the primitive quotient before applying (5). Exact arithmetic verifies monic integral coefficients and factorization; 50-digit roots provide diagnostics only.

The first two exact polynomials are

$$T^2 + 2T - 6, \quad T^2 - 2T - 4.$$

Their heights are $(1/2) \log 6$ and $\log 2$, respectively. An independent implementation reconstructs periods through 9 and uses exact rational Sturm intervals. The endpoint root at period 1 is checked by exact equality rather than by a false strict interval. Ordinary and optimized test runs agree, and all 24 hostile mutations are rejected.

7 What the next pressure must measure

Theorem 5.1 identifies a missing clock factor. Three natural repairs remain:

1. the extensive individual weight $e^{-snh(\alpha)}$;
2. a packet weight built from $\log M(\tilde{\Psi}_n) = \sum_{\alpha} \log^+ |\alpha|$;
3. discriminant, different, or ramification height, which measures conjugate interaction rather than one bounded coordinate.

The first two invite an equidistribution theorem for the complete reflection root ensemble under the full two-shift. None automatically supplies a rational-prime label or von Mangoldt amplitude.

8 Route evaluation and claim boundary

At Route A, the all-period primitive population remains analytic and exact, but the tested height pressure is a proved obstruction rather than an arithmetic bridge. We record

$$(A1_{\text{analytic}}, A2_{\text{inherited}}, A3_{\text{partial}}, A4_{\text{formal}}),$$

with overall status `ROUTE_A_EXPLORATORY`. Route B is not testable: there is no proposed Hilbert space, operator domain, self-adjoint realization, prime-power trace, or completed- ξ determinant.

No result here proves a prime correspondence, a Hilbert–Pólya realization, the Riemann hypothesis, or even a nontrivial extensive height pressure.

9 Conclusion

The complete real algebraic exhaustion of the Hénon periodic points makes a simple recurrence estimate unexpectedly decisive. Ordinary primitive coordinate heights stay bounded while the root population grows exponentially, forcing the corresponding pressure to be flat. This closes a misnormalized all-period bridge and replaces it with a sharper task: identify and control an extensive packet observable before attempting any arithmetic trace.

References

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